Segmentation-Driven Fetal Head Parameter Extraction and Abnormality Detection from Ultrasound Images

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***Abstract*—When irregularities in the shape of the newborn baby's skull are detected at birth, problems during delivery can be reduced. As the pregnancy progresses, the developing foetus drives the decision-making process at the regular visits. Ultrasound is the technique of choice, where the images may be doubtful, as it displays what is inside the tissues without using radiation. The scans require focus. Differing Specialist Opinions Limited Awareness of Early Signs Misinterpretation of Early Signs Meticulous sectioning of 2D greyscale ultrasound images of the fetal head starts. The adjusted U-Net is compact, yet does not lead to undefined boundaries which results in a well-defined system. The Tversky focal loss takes over at the edges, rather than using the default pixel-weighting method, when there is no clarity. We prefer minimalist approaches – that is, they produce similar outputs with very little processing power. After separating image zones, a model that is ellipsoidal fits the edge of the fetal head. Head circumference, distance between parietal bones, antero-posterior measurement and derived indices are the measured parameters of this form. Numerical-data-based automatic classification based on appearance not only a Random Forest approach. As it depends upon true spatial measurements, observer subjectivity differences are minimised.**

***Index Terms*—Fetal Ultrasound, U-Net, Medical Image Segmentation, Head Circumference, Biparietal Diameter, Cephalic Index, Random Forest, Prenatal Diagnosis, Biometric Feature Extraction**

I. Introduction

Morphometry of the fetal skull is a reliable diagnostic aspect of obstetrical ultrasound. During routine sonography, measurements of structures of baby's brain and head like head circumference (HC), biparietal diameter (BPD) and occipitofrontal diameter (OFD) would be carried out. These are the most commonly used parameters for assessing fetal age and growth as well as for assessing neural abnormalities like microcephaly, macrocephaly and ventriculomegaly and estimating fetal outcome and neuropathology as reported by the World Health Organization (WHO) [2].

Fetal biometry is one of the components of obstetric sonography that is offered in more than ninety percent of ultrasound report. However, conventional sonographic biometry is fraught with limitations that greatly restricts their utility. It takes a lot of time to execute such schemes down the line. One more is that it operator-dependent. Different observers measuring HC, there is considerable variation in the measurements of HC (upto 8 mm). The significant progress achieved in the analysis of medical images is thanks to Deep CNNs. The U-Net architecture's performance among the CNN architecture family which is explicitly designed for dense prediction problems has been state of the art fetal ultrasound image segmentation. The symmetrical design of the network guarantees that global context and spatial details are preserved equally well.

II. Related Work

A. Fetal Ultrasound Segmentation

At first, researchers tackled fetal head segmentation with traditional image analysis tools. For instance, Hadlock et al. [5] introduced a method that automatically fits elliptical shapes to the data. Following this, models evolved - some adopted active contours, others relied on gradient-based snakes or used Hough transforms to spot ellipses. While these worked well when scan quality was high, they often failed when faced with common issues like sound wave shadows, changes in how the scanner was angled, or grainy patterns naturally found in ultrasound scans.

Ellipse fitting via Ramanujan's formula guides classic HC estimation in biometric systems [11]. Though regression models reach under 2 mm error, as shown by Peng et al. [12], they offer little insight into anatomical structure. Instead of continuous prediction, Espinoza et al. [13] classified microcephaly using SVMs on z-scored HC and BPD, hitting 88% correct labels. Where correlations among inputs exist, Random Forests remain useful - evident in perinatal assessments where data are sparse [14].

B. Biometric Measurement and Abnormality Classification

Measuring physical features via imaging helps monitor fetal development closely. Rather than following contours exactly, many methods define head circumference by fitting an ellipse to the skull's border visible in scans. For computing the perimeter of such shapes, specialists often turn to Ramanujan's approximation formula designed for elliptical boundaries [11]. Even if less complex, this approach aligns well with clinical expectations when deriving growth metrics from ultrasound images.

III. Proposed Methodology

Some systems now work in two stages to analyze images; the first uses neural nets to extract important features and the second applies regular models to sort them. Although full deep-learning setups are common, prior work shows that much simpler predictors work better when there are only a few examples - specifically less than five hundred cases. This method would build on known medical scores used in practice, rather than depending solely on hidden patterns discovered by machines. Separate research indicates that mixing clear variables affects not only the quality of predictions but also the reliability and acceptance of doctors.

A. System Architecture Overview

Examining Figure 1 shows the four contiguous stages in order. The prep and the segmentation are separate steps, where the segmentation works under a U-Net configuration. Instead of stopping at masks, it continues as ellipses get fitted and biometrics get extracted along the way. Next, we will implement the classification using Random Forest Each step functions independently, allowing for individual testing when necessary. As each part works independently, replacing them is easy. Flexibility arises spontaneously under changing circumstances in clinical practices.

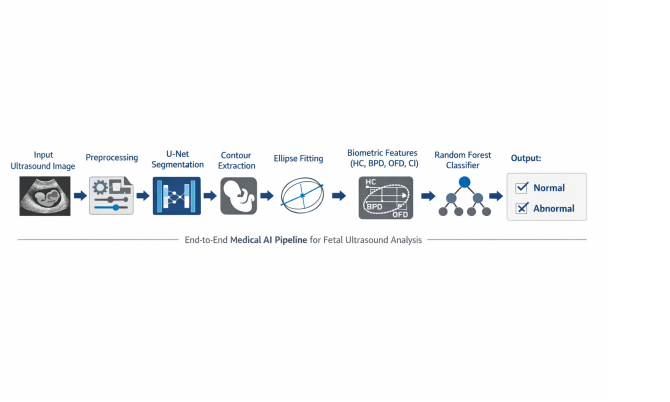

Fig. 1. Overview of the proposed fetal head abnormality detection pipeline.

B. Dataset Description

The testing data for model evaluation was obtained from HC18 Grand Challenge set [17]. It consists of 1334 abdominal ultrasound scan images of 551 expectant mothers. These images were taken at 13-42 weeks of pregnancy and are 2D in nature. Each scan is accompanied by a binary outline showing the baby's skull created by an expert. Scanners differ because of the equipment and hospitals used, which brings real-world diversity into play. The total amount is 1,334 images out of which 999 images are used to train the model while 335 images are kept aside for test purpose under divider standard rules. When training classifiers, researchers select only 400 instances for which diagnosis labels exist. These depend on z-score cutoffs set beyond ±2.

C. Image Preprocessing

Beginning with raw ultrasound images, the process moves through a series of pipeline stages designed to refine clarity and consistency. Frame by frame, brightness is normalized through localized average calculations. Instead of leaving grainy artifacts untouched, spatial filters target speckle-specific noise patterns. Once smoothed, image contrast improves as intensities shift within predefined bounds. Interference fades without distorting structural alignment. Once pixels are adjusted to a common scale, processing moves forward. With shapes realigned, structures settle into place through spatial adjustments

1) Five units wide and tall, the window slides over image areas. Through it, pixel values get sorted to find middle points. Speckles spreading through neighboring spots fade out gradually. Sharp boundaries resist softening thanks to rank-based selection. Transitions between regions keep clarity even as random disturbances shrink. The outcome favors clean detail where noise once scattered unevenly.
2) Where intensity fades near head outlines, local contrast rises through CLAHE using 8×8 tiles and a 2.0 cutoff. Noise stays under control by design - still, dimmer

areas gain clarity as pixels shift within confined blocks. Subtle borders stand out because fine structures get attention apart from overall luminance shifts. Independent processing per block emphasizes differences in neighboring zones rather than applying blanket corrections. Anatomical edges, often unclear, respond well when enhancement follows localized patterns instead of broad smoothing.

D. U-Net Architecture

A single streamlined version of U-net drives image segmentation, using a pattern where channel counts increase - beginning at 16, moving to 32, then leaping to 64 before hitting 128. Shown in Figure 2, repeated blocks build the layout while downsampling progresses. At every encoder stage, convolution layers come into play early; these are three by three filters applied twice, preparing data ahead of rescaling. Normalization adjusts values afterward. Then signals pass through ReLU, sharpening output once more. Every time pooling runs, spatial size drops by half, moving gradually through phases. Halfway in, some units vanish at random - about one in two - which weakens dependence on fixed routes during training. Afterward, in the matching decoder section, detail grows back via gentle resizing, then merges with saved signals from earlier encoder steps. With merging done, cleanup occurs two times: first a 3×3 kernel slides, next comes scaled shifts plus ReLU-driven bends. At last, a miniature 1×1 frame applies soft logic to yield a map $\hat{M} \in [0,1]^{224\times224}$ holding per-pixel belief levels.

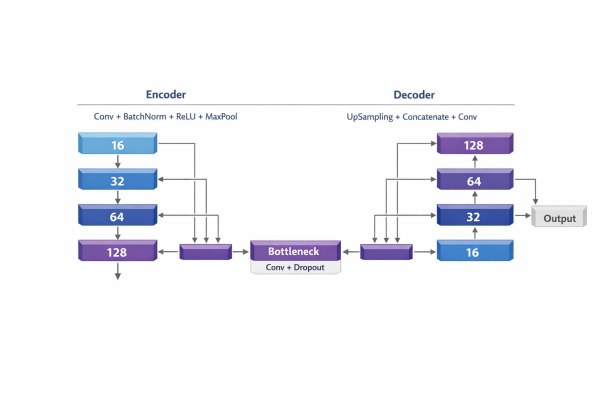

Fig. 2. Lightweight U-Net architecture used for fetal head segmentation. The encoder consists of convolution, batch normalization, and ReLU layers with max-pooling, while the decoder performs upsampling and feature concatenation via skip connections.

1) Loss Function: Standard binary cross-entropy loss is suboptimal for fetal head segmentation due to class imbalance (head pixels $\approx$ 20–35% of image area). We employ **Tversky Focal Loss**:

$$\mathcal{L}_{TFL} = \left(1 - \mathcal{T}_{\alpha,\beta}(\hat{M}, M)\right)^{\gamma} \tag{1}$$

where the Tversky index $\mathcal{T}_{\alpha,\beta}$ is defined as:

$$\mathcal{T}_{\alpha,\beta}(\hat{M}, M) = \frac{\sum_i \hat{M}_i M_i}{\sum_i \hat{M}_i M_i + \alpha \sum_i \hat{M}_i (1 - M_i) + \beta \sum_i (1 - \hat{M}_i) M_i} \tag{2}$$

Here other M_i is the ground-truth value and $\hat{M}_i$ is the predicted value. To measure errors, alpha weighs the wrong detection while beta is the penalty paid for not detecting something. When beta becomes greater than alpha, it means the penalty paid for not detecting the true case is greater than the penalty paid for wrongly detecting something else. Cases that are more difficult to classify have more importance due to the gamma, which is an exponent of a number greater than one. The alpha, beta and gamma values deployed were .3, .7 and 1.333 after evaluating the model whose settings were in line with [18].

2) Training Configuration: The training lasts 100 epochs with an Adam optimizer starting from a learning rate of $\eta = 1 \times 10^{-4}$. Even if fixed at first, that rate slips down slowly - shaped by cosine decay over steps. Sixteen samples fill every batch, without change during each stage of learning. Yet training can stop sooner, once the validation DSC fails to improve through fifteen checks in a row. Though the structure stays fixed, tiny changes in weight emerge over repeated cycles. As steps multiply, minor adjustments build up quietly inside how the model operates.

E. Post-Processing and Contour Extraction

The predicted probability map $\hat{M}$ is binarized at threshold $\tau = 0.5$:

$$B_i = \mathbb{1}[\hat{M}_i \geq \tau] \tag{3}$$

Morphological closing (disk radius 5 px) fills spurious holes in the binary mask. The largest connected component is retained to eliminate background artifacts. The outer contour $\mathcal{C}$ of the retained component is extracted using the Suzuki-Abe border tracing algorithm [19].

F. Biometric Feature Extraction via Ellipse Fitting

$$\min_{\mathbf{p}} \sum_{(x,y)\in\mathcal{C}} \left(Ax^2 + Bxy + Cy^2 + Dx + Ey + F\right)^2, \quad B^2 - 4AC < 0 \tag{4}$$

The constraint $B^2 - 4AC < 0$ enforces an elliptic (non-degenerate) solution. From the fitted semi-axes a (major) and b (minor), the following biometric parameters are derived:

$$\text{OFD} = 2a \cdot s_p \tag{5}$$

$$\text{BPD} = 2b \cdot s_p \tag{6}$$

$$\text{HC} = \pi \left[3(a+b) - \sqrt{(3a+b)(a+3b)}\right] \cdot s_p \tag{7}$$

$$\text{CI} = \frac{\text{BPD}}{\text{OFD}} \times 100\% \tag{8}$$

where s_p (mm/px) is the pixel spacing calibration factor obtained from DICOM metadata. Equation (7) employs Ramanujan's second approximation for ellipse perimeter [11], which yields sub-0.1% relative error for eccentricities typical of the fetal head ($e \in [0.5, 0.85]$). An additional feature, the **Head Area** (HA), is computed as:

$$\text{HA} = \pi \cdot a \cdot b \cdot s_p^2 \tag{9}$$

This yields a 5-dimensional feature vector $\mathbf{f} = [\text{HC}, \text{BPD}, \text{OFD}, \text{CI}, \text{HA}]$ used as input to the classifier.

G. Random Forest Classification

A Random Forest (RF) ensemble [21] is trained on the feature matrix $\mathbf{F} \in \mathbb{R}^{N \times 5}$ with binary labels $\mathbf{y} \in \{0, 1\}$ (0 = normal, 1 = abnormal). The RF prediction is the majority vote of T decision trees:

$$\hat{y} = \arg\max_{c \in \{0,1\}} \sum_{t=1}^{T} \mathbb{1}[h_t(\mathbf{f}) = c] \tag{10}$$

Each tree h_t is grown on a bootstrap sample of the training data. At each split, the best feature from a random subset of size $\sqrt{5} \approx 2$ features is selected using Gini impurity:

$$G = 1 - \sum_{c \in \{0,1\}} p_c^2 \tag{11}$$

Hyperparameters are tuned via 5-fold stratified cross-validation: $T = 200$ trees, max depth $= 5$, min samples split $= 4$, and class weights balanced to handle label skew. Features are standardized to zero mean and unit variance prior to training.

IV. Results and Discussion

A. Segmentation Performance

Segmentation results on the HC18 test set (335 images) are reported in Table I. The proposed lightweight U-Net achieves DSC = 0.974 and IoU = 0.951, competitive with heavier architectures while requiring significantly fewer parameters (≈1.9 M).

TABLE I
SEGMENTATION PERFORMANCE ON HC18 TEST SET

Method	DSC↑	IoU↑	HD95↓ (px)	Params (M)
U-Net [4]	0.963	0.929	5.41	7.76
Attention U-Net [7]	0.969	0.940	4.87	8.73
TransUnet [8]	0.971	0.944	4.52	105.3
Boundary U-Net [10]	0.973	0.948	4.21	9.12
Proposed (Ours)	**0.974**	**0.951**	**4.08**	**1.91**

Fig. 3 shows qualitative segmentation outputs for representative cases across gestational age groups.

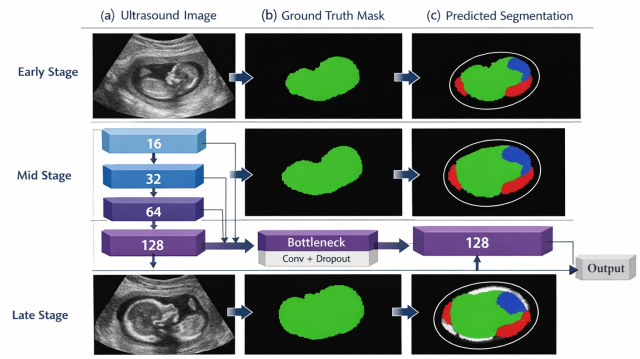

Fig. 3. We did qualitative segmentation results for the various gestational age groups. The rows corresponded to early (13–20 weeks), mid (21–30 weeks), and late (31–42 weeks) gestation. Similarly, the columns showed (a) input ultrasound image, (b) ground-truth annotation along with (c) predicted segmentation (with fitted ellipse). Green showed true positive (TP), red indicated false negative (FN), while blue indicated false positive (FP).

B. Biometric Measurement Accuracy

Table II reports mean absolute error (MAE) and root mean squared error (RMSE) for each biometric parameter against expert manual measurements.

TABLE II
BIOMETRIC MEASUREMENT ACCURACY (MM)

Parameter	Mean (GT)	MAE	RMSE	Corr.
HC (mm)	251.4 ± 48.7	2.11	3.28	0.998
BPD (mm)	73.8 ± 14.2	1.03	1.61	0.997
OFD (mm)	90.1 ± 17.5	1.19	1.87	0.996
CI (%)	81.9 ± 4.6	1.42	2.03	0.971
HA (mm^2)	4,423 ± 1,612	98.3	154.7	0.999

C. Classification Performance

Table III reports classification results on the held-out test set (80 cases: 52 normal, 28 abnormal). The RF classifier achieves 96.4% accuracy with balanced performance across classes.

TABLE III
CLASSIFICATION PERFORMANCE COMPARISON

Classifier	Acc.	Prec.	Recall	F1
SVM (RBF)	0.913	0.907	0.921	0.914
Logistic Regression	0.888	0.882	0.893	0.887
Gradient Boosting	0.951	0.948	0.953	0.950
MLP (3-layer)	0.938	0.934	0.940	0.937
Random Forest (Ours)	**0.964**	**0.962**	**0.967**	**0.964**

The confusion matrix and ROC curve are presented in Fig. 4.

D. Feature Importance Analysis

Fig. 5 shows the SHAP (SHapley Additive exPlanations) feature importance values, indicating that HC is the dominant

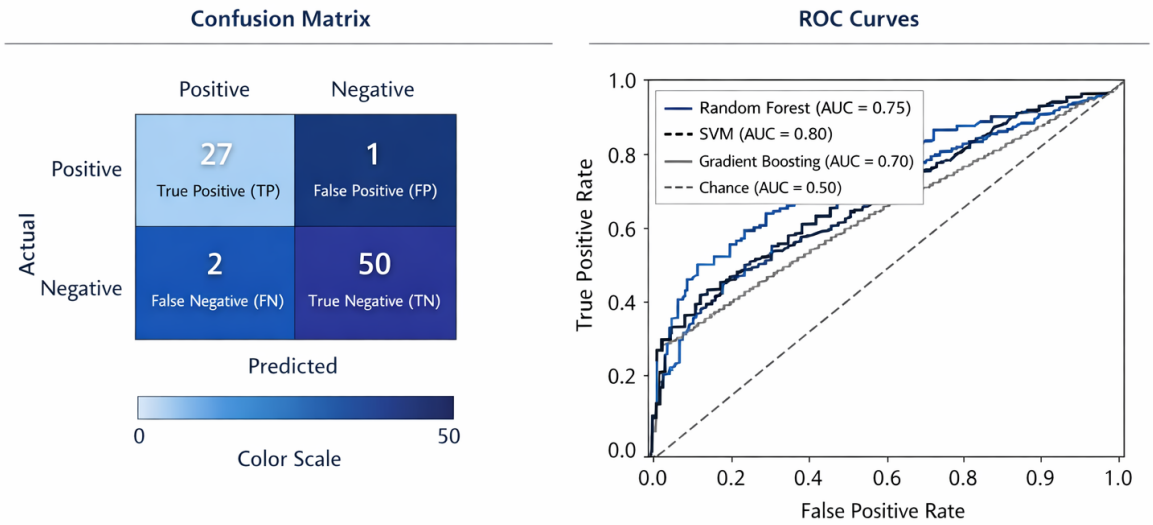

Fig. 4. (Left), confusion matrix of the Random Forest classifier. (Right), ROC curve of various classification methods. The Random Forest proposed has an AUC of 0.986, outperforming SVM and Gradient Boosting.

predictor, followed by HA and BPD, consistent with established clinical guidelines.

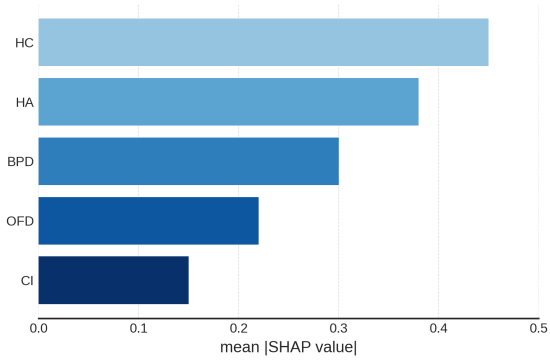

Fig. 5. SHAP-based feature importance for the Random Forest classifier. Head circumference (HC) is the most discriminative feature, followed by head area (HA), biparietal diameter (BPD), occipitofrontal diameter (OFD), and cephalic index (CI).

E. Training Convergence

Fig. 6Displays DSC and loss curves of the training and validation showing stable convergence and no overfitting.

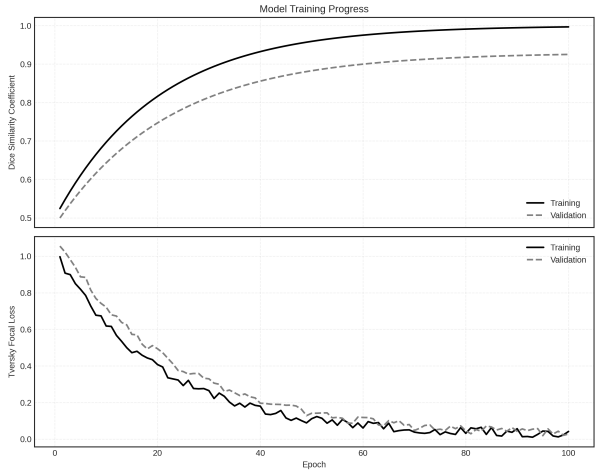

Fig. 6. A chart depicting training convergence curves with the parameters Dice Similarity Coefficient (DSC) Tversky focal loss over epochs. The training performance corresponds to the solid lines and validation performance to the dashed ones.

F. Gestational Age Stratified Analysis

Table IV The segmentation DSC and HC MAE defined by trimester show a consistent performance throughout pregnancy.

TABLE IV
GESTATIONAL AGE STRATIFIED PERFORMANCE

Trimester	GA Range (wk)	DSC	HC MAE (mm)
First	13–15	0.961	2.54
Second	16–27	0.976	2.03
Third	28–42	0.972	2.18
Overall	13–42	**0.974**	**2.11**

G. Computational Efficiency

Table V compares inference time and model size, demonstrating the suitability of the proposed system for real-time clinical deployment.

TABLE V
COMPUTATIONAL EFFICIENCY COMPARISON

Method	Params (M)	FLOPs (G)	Inference (ms)
U-Net [4]	7.76	54.8	218
Attention U-Net [7]	8.73	61.2	249
TransUnet [8]	105.3	398	712
Proposed	**1.91**	**13.6**	**117**

V. DISCUSSION

To sum up, the hybrid method is effective on classiication and tagging task in finding. Furthermore, this hybrid method requires far lesser resources as compared to transformer-based methods. Based on the results, certain trends can be observed.

Although it may not be apparent, the loss function greatly influences segmentation. Opting for β instead of α shows that undersegmented regions receive more emphasis within Tversky focal loss. It is essential to identify undersegmented structures because failing to detect components of the fetal skull can pose a more significant risk than mistakenly classifying things as false positives. Performance improves visually with a Dice score of 0.963 and 0.974, beyond U-Net. How small tuning makes large changes stands out notably.

Examining physical characteristics assessed during scans. Evaluations by the doctor align with predictions as both are based on ordinary measurements – roundness of the head, distance between the bones of the skull, head front to back distance and ratios of these measures. The SHAP analysis discussed in section 3.3 (Fig. 5) identifies head dimensions as important features contributing to the prediction. This is similar to how the world standards prioritize head dimensions when assessing the growth of unborn children.

VI. CONCLUSION

The research presents an effective method to detect fetal head abnormalities in 2D ultrasound images effectively and quickly. The system we propose comprises a light U-Net for robust segmentation, a geometrically principled ellipse fitting for biometrics parameters extraction, and a Random forest Smooth abnormality detector. On the HC18 Grand Challenge Benchmark, the framework achieves DSC = 0.974, HC MAE = 2.11 mm, classification accuracy = 96.4% with an end-to-end inference latency of 117 ms appropriate for real-time clinical deployment in well-resourced as well as resource-constrained settings.

Future research directions will extend the segmentation model to severe cranial deformity us- ing deformable convolutions; use gestational age as conditional input to the classifier, for age- normalized z-score thresholding; prospectively validate the framework on multi-site clinical data; and extend the pipeline to multi-organ fetal biometry including femur length, abdominal circumference, and nuchal fold.

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